Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Real Estate SaaS · Python · PHP · Java · PostgreSQL · Distributed Systems · AI-First · Full SDLC · AWS · Production Scale

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years owning production systems real customers and real money depend on. At Entrata (real estate SaaS), owned the full lifecycle of 20+ backend modules — from first principles design through deployment, monitoring, and production incident resolution — serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints without project managers, without hand-holding, and without the luxury of being wrong. Language-agnostic: Python, PHP, Java, TypeScript, SQL across distributed systems, event-driven architecture, and cloud infrastructure. Default to AI: built AgentProbe (open-source agent eval toolkit) and Leasel (8-agent LangGraph property operations platform) because the tooling didn't exist and the problem needed solving. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, data pipelines, and production APIs; debugging cross-stack reliability issues in data-driven features.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); Isolation Forest anomaly detection; HuggingFace fine-tuning + inference deployment; hypothesis testing and statistical validation. Workshops to 600+ students.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned production systems real customers and real money depended on** — property reservations, leasing, resident communications, financial settlement, document workflows, and compliance pipelines for 100K+ housing units. Seven years of full-lifecycle ownership: designed, built, tested, deployed, operated, and fixed when things broke. Zero compliance violations. Zero security incidents. 99.5% uptime.
- **Built fault-tolerant distributed systems from first principles** (RabbitMQ event-driven, multi-tenant PostgreSQL, AWS S3/EC2/RDS, Docker microservices) — retry logic, idempotency guarantees, concurrency-safe state transitions. Designed for correctness under pressure, not ideal conditions. 50% scalability improvement, 40% deployment time reduction, 30% query performance gain (200+ SQL optimizations via execution plan analysis, indexing, partitioning).
- **Instrumented + operated production systems** (OpenTelemetry, Prometheus, Grafana) — monitoring, alerting, dashboards; cut mean time-to-diagnose by 40%; owned every production incident through to permanent root-cause fix. Mentored 5+ engineers; 300+ code reviews; eliminated 80% of manual operational workload through Python/Bash automation.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built real estate backend modules (PHP, Python, Java, PostgreSQL); led AWS S3 migration; data pipelines across hundreds of thousands of property records.

PROJECTS

Leasel: AI-Powered Property Operations Platform PROPTECH AI LANGGRAPH

Python · LangGraph · FastAPI · RAG · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph platform automating property operations investigation - durable state, tool orchestration, fault-tolerant HITL escalation, citation-grounded RAG over lease and regulatory documents. 93.6% precision, 87.9% F1 across 1,240 residential lease PDFs. Core rule: LLM reasons, deterministic logic decides every regulated outcome. Built because the tooling didn't exist and the problem was real.

AgentProbe: Open-Source AI Eval Infrastructure CI-INTEGRATED OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; LLM-as-judge scoring; CI regression harness; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built from scratch because I needed it. Same instinct Opendoor describes: identify what needs to exist, build it.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages (agnostic): Python (8+ yrs), PHP (7 yrs production), Java (Spring Boot), JavaScript, TypeScript, SQL, PL/SQL, R, Bash — FastAPI, Django, React.js, REST APIs

Distributed Systems & Backend: Event-Driven Architecture (RabbitMQ, Kafka), Microservices, Multi-Tenant Architecture, Idempotency, Fault-Tolerant Design, Distributed Systems, PostgreSQL, Redis, Docker, Kubernetes, AWS (S3, EC2, RDS, Lambda, Step Functions)

AI & Observability: LangGraph, LangChain, RAG Pipelines, LLM Evaluation, AgentProbe, Agentic Workflows, OpenTelemetry, Prometheus, Grafana, CI/CD (GitHub Actions), Terraform, Performance Engineering